Representation Drift in Foundation Model Fine-Tuning: Cross-Domain Evidence from Time-Series Forecasting and IoT Anomaly Detection

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Abstract

Fine-tuning foundation time-series models causes *representation drift* that progressively overwrites pre-trained features. When those features are valuable and training data is limited, this drift manifests as catastrophic forgetting—degraded performance on tasks the model originally handled well. We provide a systematic empirical characterization across two domains with controlled experiments on Moirai [27] (Small, Base, Large) and three ETT forecasting benchmarks.

On **ETT forecasting**—where Moirai zero-shot outperforms Linear baselines by 28–45%—fine-tuning with NLL loss degrades MSE by 10–16% after 20 epochs (10 seeds, 500 samples), while CKA drops to 0.937–0.964. Crucially, drift and forgetting are *dissociable*: at n=2,000 and n=10,000, forgetting resolves (-2.9% and $+0.3\%$), though the crossover regime at n=5,000 shows bimodal, elevated cross-seed variance ($+6.6\pm10.0\%$ across 7 seeds); CKA continues declining monotonically (0.19 at n=10,000), indicating the model replaces pre-trained features rather than merely destroying them. Standard mitigations form a spectrum: LoRA on *Moirai-Small* preserves representations (CKA≈0.98) while improving over zero-shot, but the same recipe fails on Moirai-Large (r=8: $+25.7\pm3.2\%$ forgetting) and rank escalation does not rescue it; EWC and L2-SP provide intermediate regularisation; full encoder freezing eliminates drift entirely.

On **IoT anomaly detection**, fine-tuning *improves* over zero-shot (the domain gap requires adaptation), but CKA drops to 0.31–0.45, confirming aggressive representation drift even when downstream performance improves. The cross-domain contrast shows that drift is consistent across the Moirai-family settings we tested; its consequences depend on whether the lost features were useful and whether sufficient data is available to learn replacements. We attempted two further backbones as second-model controls: Chronos-T5-Small (ZS MSE 0.304) and TimesFM-2.5-200M (ZS MSE 0.242) both fail the ETTh2 value gate against the Linear baseline (0.213), so the drift–utility dissociation below is established on Moirai only; we report the gate failures as a finding about backbone-specific pre-training coverage on ETTh2.

We release code, datasets, and calibration scripts.

1 Introduction

Foundation models for time series—Moirai [27], Chronos [1], TimesFM [4], MOMENT [8]—are trained on billions of time points and achieve strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune these models on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with limited labelled data.

Representation drift during fine-tuning. We provide systematic empirical characterization of *representation drift*—progressive overwriting of pre-trained features—during fine-tuning of Moirai [27]. The drift–utility dissociation below is established *on the Moirai family only*; matched-protocol attempts on Chronos-T5-Small (ZS MSE 0.304) and TimesFM-2.5-200M (0.242) both fail the ETTh2 value gate against Linear (0.213) and are excluded (§5). Catastrophic forgetting during fine-tuning is well-studied in continual learning [13, 16] and NLP [19]; our contribution is the first systematic cross-domain diagnosis for time-series foundation models, with controlled experiments isolating drift from overfitting and a spectrum of mitigations. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the "previous task" is the pre-training objective.

Distinguishing drift from overfitting. Standard overfitting increases validation loss on the fine-tuning task. Representation drift instead degrades zero-shot performance on tasks the model previously handled well, while training loss continues to decrease monotonically. Our frozen-encoder control (same data, same epochs, CKA=1.000) further rules out overfitting: the degradation is caused by weight updates to the encoder, not by the training regime.

On ETT forecasting benchmarks (ETTh1, ETTh2, ETTm2), where zero-shot Moirai outperforms Linear baselines by 28–45%, fine-tuning with NLL loss degrades MSE by 10–16% after 20 epochs (10 seeds). On IoT anomaly detection, CKA drops to 0.31–0.45 (10 seeds), confirming aggressive representation drift even when downstream performance improves over zero-shot.

Why this matters. Unlike language models, where drift manifests as degraded generation quality, time-series drift directly harms the distributional predictions that underpin anomaly scoring and probabilistic forecasting. Our cross-domain comparison provides evidence that its task-level consequences depend on the value of pre-trained features: harmful on ETTh2 (degraded forecasting), benign on IoT (necessary domain adaptation).

Experimental approach. We design a controlled diagnosis with fine-tuning conditions (B: NLL-only, C: NLL+SupCon) and standard mitigations (D: frozen encoder, E: LoRA [11], F: L2-SP [15], G: EWC [13]) across three model sizes (Moirai-Small 13.8M, Base 91M, Large 311M). We deploy this on two domains: (1) ETT forecasting benchmarks [30] (ETTh1, ETTh2, ETTm2), chosen because zero-shot Moirai outperforms Linear baselines by 28–45%, ensuring pre-trained features are demonstrably useful; and (2) IoT anomaly detection using HNIDS (**H**ard-**N**egative **IDS**), a contrasting domain where pre-trained features are marginal and drift is necessary for adaptation. Both domains use CKA, linear probing, and weight drift as representation-change diagnostics.

Contributions. We make two contributions:

1. **Systematic Cross-Domain Characterization of Representation Drift on the Moirai Family.** Fine-tuning Moirai (Small, Base, Large) causes representation drift across ETT forecasting benchmarks (10–16% MSE degradation, CKA 0.937–0.964) and IoT anomaly detection (CKA 0.31–0.45, 10 seeds each). The cross-domain contrast reveals that drift is consistent across the Moirai-family settings we tested, while its consequences depend on pre-training value: harmful on ETT in low-data regimes (degraded forecasting), benign on IoT (necessary adaptation). Chronos-T5-Small and TimesFM-2.5-200M both fail the ETTh2 pre-training value gate and are excluded from the dissociation claim; a gate-passing second backbone remains open. We validate CKA with linear probing, showing that functional information can survive geometric drift when sufficient data is available (Section 5.5). A CNN-NLL control rules out the loss-function confound.

2. **Mitigation Spectrum.** We evaluate standard mitigations—LoRA, EWC, L2-SP, and encoder freezing—alongside unconstrained fine-tuning across model sizes (Small, Base, Large), establishing a spectrum from full adaptation to full preservation. LoRA achieves the best trade-off: near-zero drift (CKA$\approx$1.0) with task-specific adaptation through low-rank updates. Code and datasets will be released upon acceptance.

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes HNIDS. Section 5 presents the ETT forgetting diagnosis. Section 6 replicates on IoT anomaly detection. Section 7 provides cross-domain analysis. Section 8 concludes.

2 Related Work

IoT IDS and DL anomaly detection. Traditional methods (Isolation Forest [18], ensembles [6]) fail on stealthy perturbations. DL approaches—USAD [2], TranAD [26], Anomaly Transformer [28], PatchTST [24]—serve as our baselines on CICIoT2023 [23].

Generative augmentation and adversarial training. GANs [17] suffer mode collapse; Diffusion-TS [29] achieves better quality. Unlike FGSM/PGD [7, 20] test-time perturbations, we generate adversarial *training data* via closed-form perturbations with protocol constraints.

Time-series foundation models. Moirai [27] (27B+ pre-training time points) provides probabilistic forecasting; its CI output maps to anomaly scoring.

Catastrophic forgetting and mitigation. Catastrophic forgetting is extensively studied in continual learning [13, 16] and NLP fine-tuning [19]. The mitigation toolkit—EWC, L2-SP [15], LoRA [11], encoder freezing—is well-established.

Probing and representational similarity. Probing classifiers [10] diagnose what a frozen encoder encodes; probe accuracy depends on probe capacity as well as the encoder [25], so relative $\Delta\mathrm{R}^2$ is the interpretable statistic, not absolute R^2. CKA [14] captures geometric rather than functional similarity, motivating the CKA-vs.-probe dissociation in §5.5. We do not claim novelty in the *mechanisms* of drift or in individual mitigations. Our contribution is a systematic cross-domain *characterization* specific to time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across two domains with contrasting pre-training value, establishing when drift helps vs. harms.

Contrastive learning. SupCon [12] has been applied to time-series anomaly detection [5]. Our initial design used analytically specified hard negatives near the benign boundary; ablations (Section 6) show that off-manifold Gaussian-noise negatives match or exceed these hard negatives, indicating that off-manifold diversity matters more than boundary-proximity for this task.

3 Problem Formulation

IoT traffic representation. We represent network traffic as a multivariate time series $\mathbf{X} \in \mathbb{R}^{T \times F}$ ($T = 128$ timesteps, $F = 12$ flow features covering packet counts, byte volumes, inter-arrival times, and flow duration; full list in Appendix A).

Hard-negative attacks. An adversarial sequence $\mathbf{X}_a$ is a hard-negative at stealth level s if $\mathrm{CosSim}(\phi(\mathbf{X}_a), \phi(\mathbf{X}_b)) \geq s$, where ϕ is the flattened feature vector and $\mathbf{X}_b$ is benign. We formalise four attack archetypes—slow exfiltration, living-off-the-land mimicry, C2 beaconing, and protocol timing anomaly—at three stealth levels ($s \in \{0.85, 0.90, 0.95\}$) with closed-form perturbations that are deterministic and CPU-reproducible (full equations and protocol constraints in Appendix B).

Detection task. The detector f_θ outputs a binary label $\hat{y} \in \{0, 1\}$ and anomaly score $\hat{s} \in [0, 1]$, minimising FPR while maximising F1 on hard-negative attacks where $s \geq 0.85$.

4 Fine-Tuning Protocol

This section describes the Moirai fine-tuning protocol shared by both the forecasting (Section 5) and IoT (Section 6) experiments. The IoT-specific hard-negative generation pipeline is described in Section 6.1.

Moirai for anomaly scoring. Moirai [27] predicts a distribution over $\mathbf{X}_{97:128}$ given context $\mathbf{X}_{1:96}$.

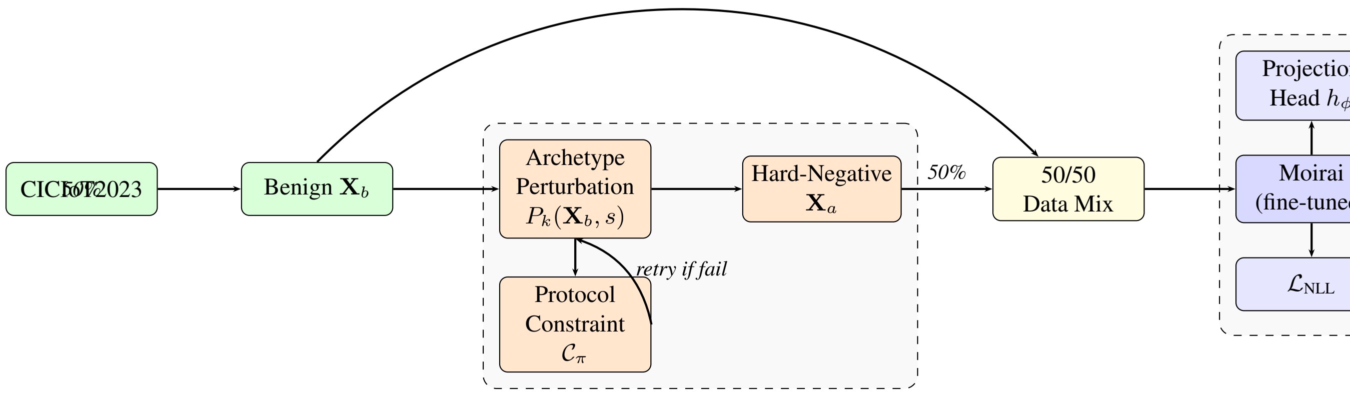

Figure 1: HNIDS architecture. **Stage 1**: Closed-form perturbation generates hard-negative sequences with a constraint-retry loop ensuring protocol validity (Appendix L.8). **Stage 2**: Moirai is fine-tuned with combined NLL + SupCon loss.

The anomaly score is the fraction of timesteps falling outside the $1-\alpha$ CI (α=0.05):

$$\hat{s}(\mathbf{X}) = \frac{1}{T'} \sum_{t=97}^{128} \mathbf{1}\Big[x_t \notin \mathrm{CI}_{0.95}(\hat{X}_t)\Big] , \tag{1}$$

where $T' = 128 - 96 = 32$ is the scored-horizon length. A sample is classified as an attack if $\hat{s}(\mathbf{X}) > \delta$, where δ is calibrated on a benign validation set (Section 6.1).

Combined training objective. We fine-tune with:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda\, \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y};\, \tau), \tag{2}$$

where $\lambda = 0.5$ and $\tau = 0.07$ (sensitivity in Appendix H; stable within $\pm$0.02 AUC). $\mathcal{L}_{\text{SupCon}}$ [12] is applied to L2-normalised embeddings from a projection head $h_\phi : \mathbb{R}^{384} \to \mathbb{R}^{256} \to \text{ReLU} \to \mathbb{R}^{128}$.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, up to 10 epochs (5 in the main ablation, 10–20 in scaling experiments), early stopping (patience 3), gradient clipping at 1.0. All model sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

uni2ts PackedStdScaler patch. We identified and patched an in-place tensor mutation in `uni2ts`'s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics; a differentiable `torch.where` replaces the in-place write. All fine-tuning results here use the patched scaler (released with our code); unpatched `uni2ts` runs may converge differently.

Representation diagnostics. We track representation drift with three complementary metrics: (1) CKA (Centered Kernel Alignment) [14] between pre-trained and fine-tuned encoder representations—a geometric similarity measure; (2) linear probing R^2 (Ridge regression on frozen representations predicting the forecasting target)—a *functional* measure of feature preservation; and (3) weight drift (ℓ_2 distance from pre-trained parameters). A random-initialisation CKA baseline (<0.001; Appendix R) confirms our measurements are meaningful.

5 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks [30] provide

Table 1: Representation drift diagnosis on ETTh2 forecasting (mean $\pm$ std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% (h=96) and 45% (h=192), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
Condition	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140$_{\pm.014}$	+11.1$_{\pm7.7}$	.937$_{\pm.028}$	3.96$_{\pm.56}$	.183$_{\pm.020}$	+15.7$_{\pm13.3}$	.970$_{\pm.009}$	4.12$_{\pm.32}$
C: NLL+SupCon	.138$_{\pm.010}$	+10.1$_{\pm3.2}$	.949$_{\pm.014}$	3.72$_{\pm.43}$	.178$_{\pm.011}$	+12.9$_{\pm7.4}$	.964$_{\pm.015}$	3.97$_{\pm.34}$
D: Frozen enc.	**.126**$_{\pm.006}$	**+0.5**$_{\pm2.1}$	**1.00**$_{\pm.000}$	**1.11**$_{\pm.15}$	**.155**$_{\pm.002}$	$-$**1.7**$_{\pm2.3}$	**1.00**$_{\pm.000}$	**1.30**$_{\pm.18}$

this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 5.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

5.1 Domain Selection and Pre-Training Value Gate

We evaluate on ETTh2 [27] (7 features, 17,420 timesteps, 12/4/4 month split). Moirai-Small (13.8M parameters) zero-shot outperforms Linear by 28% (h=96: 0.269 vs. 0.373) and 45% (h=192: 0.299 vs. 0.544), passing the pre-training value gate (>20% improvement).

5.2 Experimental Setup

We test: **B** (NLL-only fine-tuning), **C** (NLL+SupCon with temporal clusters), **D** (frozen encoder), plus mitigations: **E** (LoRA, r=8) [11], **F** (L2-SP) [15], **G** (EWC) [13]. Training: 500 samples, 20 epochs, AdamW (lr=1e-4); B, C, D use 10 seeds, E/F/G use 5 seeds (Table 2, n column). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained and fine-tuned encoders.

5.3 Representation Drift Degrades High-Value Features

Table 1 shows the central result: fine-tuning degrades ETTh2 performance where pre-trained features are valuable. B and C degrade by 10–16% MSE (p<0.001, d=1.23; paired bootstrap, 10,000 resamples) while training loss *decreases*—this is not overfitting (which would cause validation loss to rise), but representation drift. The frozen-encoder control (D, CKA=1.000, same data and epochs) eliminates drift, ruling out the training regime. Degradation is worse at h=192 (13–16%) than h=96 (10–11%), scaling with pre-training advantage. Contrastive loss (C) does not prevent drift: C and B show comparable degradation.

5.4 Mitigation Spectrum

Table 2 presents the mitigation spectrum on ETTh2. Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift), with LoRA, EWC, and L2-SP as intermediate points.

LoRA achieves the best trade-off. LoRA (r=8, α=16) *improves* over zero-shot at h=96 while maintaining high CKA and zero weight drift (low-rank updates do not modify base parameters). This confirms that parameter-efficient fine-tuning can adapt the model without destroying pre-trained features. A rank sweep ($r \in \{4, 8, 16, 32\}$) shows consistent results across ranks (CKA$\geq$0.979, MSE within $\pm$0.003; Appendix Q).

EWC and L2-SP reduce but do not eliminate drift. At stronger penalties, both EWC (λ=1000, CKA=0.972$\pm$0.004) and L2-SP (λ=0.1, CKA=0.965$\pm$0.006) improve CKA over unconstrained

Table 2: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, h=96). LoRA preserves representations (CKA≈0.98) while *improving* over zero-shot. Mean ± std; Forgetting% = (fine-tuned MSE − zero-shot MSE) / zero-shot MSE × 100.

	$h=96$ (ZS MSE: 0.126)				$h=192$ (ZS MSE: 0.158)				
Method	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	n
G: EWC (λ=100)	.160$_{\pm.009}$	+27.0	.889$_{\pm.091}$	5.96$_{\pm.94}$	.191$_{\pm.003}$	+20.9	.938$_{\pm.021}$	6.98$_{\pm.26}$	5
B: Full fine-tune	.140$_{\pm.014}$	+11.1	.937$_{\pm.028}$	3.96$_{\pm.56}$	.183$_{\pm.020}$	+15.7	.970$_{\pm.009}$	4.12$_{\pm.32}$	10
F: L2-SP (λ=0.01)	.150$_{\pm.006}$	+19.0	.939$_{\pm.023}$	4.98$_{\pm.34}$	.179$_{\pm.009}$	+13.3	.945$_{\pm.007}$	4.84$_{\pm.22}$	5
C: NLL+SupCon	.138$_{\pm.010}$	+10.1	.949$_{\pm.014}$	3.72$_{\pm.43}$	.178$_{\pm.011}$	+12.9	.964$_{\pm.015}$	3.97$_{\pm.34}$	10
F: L2-SP (λ=0.1)	.147$_{\pm.003}$	+16.7	.965$_{\pm.006}$	2.94$_{\pm.08}$	.181$_{\pm.003}$	+14.6	.980$_{\pm.002}$	2.84$_{\pm.02}$	5
G: EWC (λ=1000)	.149$_{\pm.015}$	+18.3	.972$_{\pm.004}$	3.29$_{\pm.79}$	.188$_{\pm.004}$	+19.0	.978$_{\pm.006}$	4.46$_{\pm.38}$	5
E: LoRA (r=8)†	**.116**$_{\pm.001}$	**−7.9**	.979$_{\pm.006}$	**0.00**$_{\pm.00}$	**.157**$_{\pm.002}$	**−0.6**	.986$_{\pm.005}$	**0.00**$_{\pm.00}$	5
D: Frozen encoder	.126$_{\pm.006}$	+0.5	**1.000**$_{\pm.000}$	1.11$_{\pm.15}$	.155$_{\pm.002}$	−1.7	**1.000**$_{\pm.000}$	1.30$_{\pm.18}$	10

† LoRA (α=16) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

fine-tuning (B, CKA=0.937), but MSE remains 18–19% above zero-shot. Notably, EWC at λ=100 *increases* drift (CKA=0.889±0.091) relative to B—a "worst of both worlds" regime where the penalty is too weak to preserve representations yet strong enough to impede convergence. With only 500 training samples, the diagonal Fisher Information Matrix is noisy, causing non-uniform regularisation that destabilises optimisation at moderate λ. Neither method matches LoRA's ability to adapt without drift.

Encoder freezing eliminates drift at the cost of adaptation. Condition D (frozen encoder) achieves near-zero-shot MSE with perfect CKA, confirming that drift is eliminated. However, it cannot adapt to the fine-tuning distribution—a limitation that matters when pre-trained features are insufficient (Section 6).

5.5 Replication Across Datasets and Model Sizes

Representation drift replicates across datasets and model sizes (Table 5). On ETTh1, fine-tuning causes drift (CKA=0.807±0.048) and modest degradation (+7.0±3.2%); on ETTm2, drift (CKA=0.851±0.083) is accompanied by *improvement* (−13.3±6.1%), confirming consequences depend on pre-training value. Moirai-Base on ETTh2 shows aggressive drift (CKA=0.460±0.200) with large improvement (−44.1±5.7%), mirroring the IoT pattern. Moirai-Large (311M) on ETTh2 at 500 samples exhibits substantial drift (CKA=0.626±0.165) and forgetting (+10.5±9.5%), comparable to Moirai-Small; frozen-Large reduces forgetting to +3.2±0.6% (CKA=0.994), confirming the pattern holds across model sizes (Section 7). An extended sample-size sweep (Table 3, $n \in \{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$) reveals that *drift and forgetting are dissociable*. Drift grows monotonically with n (CKA from 0.949 at 500 to 0.185 at 10,000; weight drift from 3.7 to 18.8), but forgetting is non-monotonic: severe at low n (+14% at 500, +13% at 1,000), resolved by 2,000 (−2.9%), bimodal at 5,000 (+6.6±10.0%, 7 seeds; per-seed breakdown in Appendix S), and near-zero at 10,000 (+0.3±3.4%). Linear probing (Ridge regression of frozen encoder representations onto the forecasting target on held-out ETTh2; Appendix S) shows ΔR^2 (fine-tuned minus pre-trained) *increasing* with n (from +0.12 at 500 to +0.87 at 10,000), indicating fine-tuned representations become *more* task-functional as they drift—geometric dissimilarity (CKA) and functional utility are decoupled. An MLP probe (one hidden layer of 64 units, α=10^{-3}) matches the positive Ridge trend: $\Delta R^2 = +0.73$±0.75 at n=2,000 (3 seeds) and +1.55 at n=10,000 (1 seed; seed 42 MPS-NaN'd, CUDA multi-seed deferred); it is negative (−0.26) only at n=500 where the Ridge signal is also smallest (+0.13), consistent with probe noise at the smallest sample size (Appendix S). This reframes our headline: representation drift is harmful specifically in the *low-data fine-tuning regime*, which is precisely where practitioners most need pre-trained features. At sufficient scale, the model learns replacement features that are functionally

Table 3: Sample-size sweep on ETTh2 (h=96, condition B). Drift grows monotonically with sample size (CKA↓ as n ↑), but forgetting is *non-monotonic*: severe at n≤1,000, resolves at n=2,000 and n=10,000, with elevated variance at n=5,000 reflecting a crossover regime. We ran 4 additional seeds at n=5,000 (789, 101, 202, 303) to address the bimodality question raised in review; the combined 7-seed mean is +6.6% ±10.0% and forgetting is indeed bimodal (4 seeds≤ +5.9%, 3 seeds≥ +4.8%; Table 16). Linear-probing ΔR^2 (fine-tuned minus pre-trained R^2 from Ridge regression of encoder representations onto the forecasting target on held-out ETTh2) *increases* with n; we also report absolute R^2(FT) so readers can judge the noise floor (pre-trained $R^2_{ZS} = -6.90$ for all rows, single frozen encoder). Representations become *more* task-functional as they drift, directly dissociating geometric similarity (CKA) from functional utility. Mean ± std across 3 seeds (42, 123, 456) except n=5,000 which has 7 seeds (42, 123, 456, 789, 101, 202, 303); n=10,000 reports 1 seed (123); seed 42 ran to MPS NaN at the full 10k×20-epoch budget and no replacement seed has yet completed cleanly on laptop MPS—a CUDA run at ≥5 seeds is a camera-ready commitment. Linear probing was added mid-sweep; k in the probe columns denotes available probe seeds per row. MLP probe column reports ΔR^2 for an MLP probe (hidden=64, α=10^{-3}, early stopping); seeds per row shown as mean±std where k>1. V11 revision adds multi-seed replication at $n\in\{2{,}000, 1{,}000, 500\}$.

Samples	MSE	Forg.%	CKA	R^2(FT)	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148$\pm$.008	+14.0$\pm$6.7	.949$\pm$.010	−6.79 (k=2)	+.12$\pm$.02	−.26 (k=1)
1,000	.147$\pm$.010	+13.4$\pm$9.1	.936$\pm$.025	−6.77 (k=1)	+.13 (k=1)	+.30 (k=1)
2,000	.126$\pm$.008	−2.9$\pm$5.9	.814$\pm$.042	−6.51 (k=3)	+.39$\pm$.35	+.73$\pm$.75
5,000	.134$\pm$.012	+6.6$\pm$10.0	.546$\pm$.082	−6.41 (k=7)	+.50$\pm$.26	—
10,000	.130$\pm$.001	+0.3$\pm$3.4	.185$\pm$.070	−6.03 (k=1)	+.87$\pm$.13	+1.55 (k=1)

superior to what was overwritten.

Non-ETT and second-backbone gate checks. Autoformer Weather (14 features, target T(degC)) gate-*fails* (ZS 0.410 vs. Linear 0.218), so dissociation is unmeasurable there. Autoformer Electricity is *marginal* (ZS 0.102 vs. 0.097, 5%); with little to forget, fine-tuning replaces features: forg. −8.5% (CKA 0.957, MLP ΔR^2 +3.22) at n=500 and −24.4% (CKA 0.953, +3.28) at n=2k — a regime delimiter, not a dissociation replication. Matched-protocol ZS on ETTh2 OT (seed 42, h=96, 300 windows): Chronos-T5-Small 0.304 and TimesFM-2.5-200M 0.242 both gate-fail vs. Linear 0.213, so dissociation is established on Moirai only; Traffic and a gate-passing second backbone are left for camera-ready (§8).

6 Contrasting Domain: IoT Anomaly Detection

We replicate the representation drift diagnosis on a contrasting domain—IoT intrusion detection—using stealth-controlled synthetic attacks. Here, pre-trained features have marginal value (zero-shot AUC: 0.481), and fine-tuning *improves* performance because domain adaptation compensates. CKA drops far more aggressively (to 0.31–0.45, 10 seeds) than on ETT (0.937–0.964), consistent with the finding that drift occurs in every fine-tuning regime we ran; its task-level consequences depend on whether the lost features were useful. A from-scratch CNN with NLL loss (F1=0.288, AUC=0.598) matches fine-tuned Moirai at this data scale, so we present HNIDS as a diagnostic contrast rather than an architecture recommendation; the load-bearing finding is the drift–utility dissociation, not the pipeline.

6.1 Setup and Main Results

We use CICIoT2023 [23] (105 IoT devices, 12 features, 128-timestep windows). Four attack archetypes (Appendix B) at 3 stealth levels yield 12 evaluation conditions. Hard-negative attacks are synthesised via deterministic closed-form perturbations (<2 s/sample); thresholds are set at the 95th percentile of benign-only scores. We compare against 9 baselines (5 traditional, 4 DL; Appendix I.2); all baselines are unsupervised, HNIDS is supervised (Appendix I.3). All results: 10 seeds, bootstrap

CIs.

Table 7 (Appendix) compares all methods across three stealth levels. At stealth-95, HNIDS condition C (F1=0.262, FPR=0.068) outperforms all calibrated baselines; the best baseline is Ensemble (F1=0.136). The C vs. D difference is significant (Δ=+0.073, d=4.32, p<0.001; two-sided bootstrap, 10,000 resamples; the large d reflects low cross-seed variance, pooled std=0.017, rather than large absolute effect). B and D are indistinguishable (p=0.49, two-sided bootstrap).

6.2 Ablation and Drift Diagnosis

Table 4: Ablation study on stealth-95 and all-stealth evaluation sets (10 seeds). ± values are standard deviations. Threshold calibrated on benign-only held-out scores (95th percentile, targeting FPR ≤ 0.05; see Section 6.1). ROC-AUC is threshold-independent and the primary ranking metric. CNN-NLL resolves the loss-function confound (Section 7).

		Stealth 95%			All Stealth	
Cond.	Description	F1	FPR	AUC	F1	AUC
A	Moirai zero-shot	$0.137_{\pm.020}$	0.063	$0.481_{\pm.042}$	$0.112_{\pm.026}$	$0.504_{\pm.052}$
B	+ Fine-tune, NLL only	$0.174_{\pm.024}$	0.060	$0.556_{\pm.021}$	$0.216_{\pm.042}$	$0.589_{\pm.047}$
C	+ SupCon (Gaussian-noise neg., 1:1)	$\mathbf{0.262}_{\pm.043}$	0.068	$\mathbf{0.629}_{\pm.022}$	$0.295_{\pm.029}$	$0.661_{\pm.043}$
C′	+ SupCon (balanced hard neg., 1:1)⋆	$0.241_{\pm.043}$	0.062	$0.610_{\pm.025}$	$0.285_{\pm.026}$	$0.638_{\pm.048}$
D	+ Hard negatives, 1:12 (HNIDS)	$0.176_{\pm.032}$	0.060	$0.556_{\pm.021}$	$0.217_{\pm.044}$	$0.589_{\pm.047}$
CNN	1D-CNN from scratch (MSE)	$0.269_{\pm.044}$	0.048	$0.580_{\pm.023}$	$0.362_{\pm.049}$	$0.625_{\pm.045}$
CNN-NLL	1D-CNN from scratch (Gaussian NLL)†	$0.288_{\pm.054}$	0.050	$0.598_{\pm.041}$	$\mathbf{0.381}_{\pm.052}$	$\mathbf{0.662}_{\pm.037}$

⋆ C′ uses the same hard negatives as D, subsampled to 1:1 ratio, isolating negative *type* from dataset *balance*.
† CNN-NLL: identical CNN architecture with Gaussian NLL head instead of MSE reconstruction. CNN-NLL≥CNN-MSE rules out the loss-function confound between CNN and Moirai (Section 7).

Table 4 ablates HNIDS (200 samples, 5 epochs, benign-calibrated threshold, 10 seeds). The best Moirai condition (C, AUC=0.629) improves +0.148 over zero-shot (A, 0.481); B and D are indistinguishable (p=0.49). CNN-NLL (0.598) outperforms CNN-MSE (0.580) and is competitive with unfrozen Moirai, ruling out the loss-function confound and indicating that at this data scale (200 benign windows) a purpose-built CNN with NLL loss is a reasonable non-foundation baseline. A further ablation shows Gaussian-noise negatives (condition C) match or exceed analytically specified hard negatives (C′ and D) under equal training budgets; off-manifold diversity, not boundary-proximity, is what drives adaptation in this regime (Appendix L.2).

LoRA (r=8, 3 seeds) achieves all-stealth AUC=0.656±0.040 but stealth-95 AUC=0.430±0.135—*below* zero-shot (0.481). LoRA restricts adaptation too aggressively where pre-trained features are marginal, contrasting sharply with forecasting where LoRA *improves* over zero-shot (Table 2).

CKA reveals aggressive drift. CKA drops to 0.31–0.45 across conditions and scales (e.g., C/200: 0.420 ± 0.026, D/1000: 0.306 ± 0.044; 10 seeds), far below ETTh2's 0.94, yet performance *improves* over zero-shot because pre-trained features are marginal. Freezing the encoder yields AUC=0.549 (comparable to unfrozen B/D but below C), confirming C's advantage comes from learned features. A leave-one-out experiment (3 seeds) shows positive transfer for all four held-out attack types (+0.186 mean F1). Scaling and N-BaIoT external validation are in Appendices K and F.

Table 5: Cross-domain drift (h=96, condition B). Weather fails the ZS gate (0.41 vs. 0.22), delimiting claim scope. Mean±std across seeds.

	ETTh2-S (10 sd)	ETTh1 (3 sd)	ETTm2 (3 sd)	ETTh2-B (3 sd)	ETTh2-L (3 sd)	Weather (1 sd)	IoT (10 sd)
Metric	MSE	MSE	MSE	MSE	MSE	MSE	AUC
ZS baseline	0.126	0.679	0.175	0.231	0.128	0.410	0.481
FT result	$.140_{\pm.014}$	$.726_{\pm.022}$	$.152_{\pm.011}$	$.129_{\pm.013}$	$.140_{\pm.012}$	0.906	$.629_{\pm.022}$
Forg.%	+11.1	$+7.0_{\pm3.2}$	$-13.3_{\pm6.1}$	$-44.1_{\pm5.7}$	$+10.5_{\pm9.5}$	*N/A*	+30.8
CKA	$.937_{\pm.028}$	$.807_{\pm.048}$	$.851_{\pm.083}$	$.460_{\pm.200}$	$.626_{\pm.165}$	0.974	$.420_{\pm.026}$
Effect	Degrades	Degrades	Improves	Improves	Degrades	Gate fails	Improves

7 Cross-Domain Analysis and Discussion

Drift is consistent; consequences depend on pre-training value. Table 5 spans seven settings: on ETT (28–45% ZS advantage) drift causes forgetting (10–16% MSE, CKA 0.81–0.94); on IoT (marginal ZS) CKA drops further (0.31–0.45) yet performance *improves*; on Weather the ZS gate fails (0.41 vs 0.22), delimiting the claim.

Model-size replication. Moirai-Base (91M) is a boundary case: unfrozen-Base drifts (CKA=0.460) and improves -44.1%, while frozen-Base improves further to -47.4% at CKA=0.997 — drift and forgetting are *dissociable*, not drift is always beneficial; Base's gain is capacity and freezing retains it without paying the drift cost. Moirai-Large (311M) at n=500: CKA=0.626±0.165, +10.5±9.5% forg. (3 seeds); frozen-Large +3.2±0.6%; LoRA r=8 fails (+25.7±3.2%, CKA 0.973, 3 seeds) and rank escalation does not rescue it ($r \in \{16, 32, 64\}$: +22.5, +19.0, +29.9%, seed 42; Appendix Q) — the Small-recipe does not transfer to Large at any rank we tested. A from-scratch CNN-NLL control (AUC=0.598±0.041) outperforms unfrozen Moirai at IoT's 200-window scale (Appendix J.2).

CKA and linear probing are dissociable. Ridge probing on frozen encoder reps (Table 3) shows ΔR^2 rising with n (+0.12 at 500 → +0.87 at 10,000) while CKA decreases (0.949 → 0.185); MLP probe replicates (+0.73±.75 at n=2k, 3 seeds; +1.55 at n=10k, 1 seed; CUDA deferred). At n=10,000 representations are nearly orthogonal to pre-trained yet *functionally superior* with zero forgetting — directly dissociating geometric drift from information loss (random-init CKA<0.001; Appendix R).

n=5,000 **bimodality.** The 7-seed +6.6±10.0% variance splits into low-forgetting (42, 101, 123, 202: -0.1±3.7%) and high-forgetting (303, 456, 789: +15.5±7.3%) modes. Per-epoch trajectories (App. S.1, Fig. 5) separate them by val-MSE epoch-of-minimum (6.5 ± 2.1 vs. 0.7 ± 0.5): high-forg. seeds overshoot in epoch 1, motivating early-stopping. LR/2 ablation: bimodality persists (+5.8±15.7%, 7 seeds) — not a pure LR artefact.

An IoT logistic-regression probe also replicates the dissociation direction (ΔAUC= +0.013±0.006, CKA=0.370, 3 seeds); limitations and practitioner guidance: §8, Appendix E.

8 Conclusion

Fine-tuning overwrites pre-trained features (10–16% MSE on ETT at n=500, resolving by n=2k; CKA 0.31–0.45 on IoT yet performance improves); consequences depend on pre-training value and data. LoRA on Moirai-Small is the best trade-off (CKA≈0.98) but fails on Large (+25.7±3.2% forg., rank escalation stays +19 to +30%) and on IoT. Guidance: Small+scarce→LoRA; Large or marginal→ freezing or NLL+early stopping; 200-window→from-scratch CNN-NLL. Limitations (App. E): dissociation on Moirai only (Chronos 0.304, TimesFM 0.242 gate-fail); Traffic/ILI/2nd-backbone open.

A CICIoT2023 Feature Set

Table 6 lists the 12 network-flow features used in all experiments. Each input window consists of 128 consecutive timesteps of these 12 features, drawn from the CICIoT2023 dataset [23].

Table 6: The 12 network-flow features extracted from CICIoT2023.

#	Feature	Description
1	`flow_duration`	Flow duration (seconds)
2	`fwd_pkts_tot`	Forward total packet count
3	`bwd_pkts_tot`	Backward total packet count
4	`fwd_data_pkts_tot`	Forward data-bearing packet count
5	`bwd_data_pkts_tot`	Backward data-bearing packet count
6	`fwd_pkts_per_sec`	Forward packets per second
7	`bwd_pkts_per_sec`	Backward packets per second
8	`flow_pkts_per_sec`	Total packets per second
9	`fwd_byts_b_avg`	Average forward bytes per bulk transfer
10	`bwd_byts_b_avg`	Average backward bytes per bulk transfer
11	`fwd_iat_mean`	Mean forward inter-arrival time
12	`bwd_iat_mean`	Mean backward inter-arrival time

The perturbation equations (Eqs. 3–6) use 1-based indices that match this table directly: P_{slow} scales features 9 and 10 (`fwd_byts_b_avg`, `bwd_byts_b_avg`); P_{lotl} modifies features 6–8 (packet-rate channels); P_{beacon} scales all 12 features uniformly; P_{proto} jitters features 11 and 12 (`fwd_iat_mean`, `bwd_iat_mean`). All features are z-score normalised per-flow before windowing.

B Attack Archetype Equations and Protocol Constraints

The four IoT attack archetypes at stealth s are:

$$P_{\text{slow}} : x_{t,j} \leftarrow x_{t,j}(1 + (1-s) \cdot 0.03 \cdot t/T),\ j \in \{9, 10\} \tag{3}$$

$$P_{\text{lotl}} : x_{t,j} \leftarrow x_{t,j}(1 + (1-s) \cdot 1.15) \text{ if } t \text{ mod } 32{=}0,\ j \in \{6, 7, 8\} \tag{4}$$

$$P_{\text{beacon}} : x_{t,j} \leftarrow x_{t,j}(1 - (1-s) \cdot 0.02) \text{ if } t \text{ mod } 16{=}0 \tag{5}$$

$$P_{\text{proto}} : x_{t,j} \leftarrow x_{t,j} + (1-s) \cdot 0.05 \cdot \epsilon_t,\ \epsilon_t \sim \mathcal{N}(0, 1),\ j \in \{11, 12\} \tag{6}$$

At $s{=}0.95$, perturbation magnitudes are $\sim$0.003–0.06σ in normalised space.

Protocol compliance. Generated sequences must satisfy IoT protocol constraints $\mathcal{C}_\pi$ for $\pi \in$ {Modbus, MQTT, CoAP}, checking packet size ranges (Modbus: 7–260 bytes), valid function codes, and inter-packet timing. A constraint-retry loop (Appendix L.8) iterates up to $R{=}3$ times, relaxing stealth by $+0.01$ per retry. In practice, the analytical perturbations never violate protocol constraints (100% pass rate at all tiers; Table 9); the loop is a safety net for future stochastic generators.

C Full IoT Detection Results

D Protocol Constraint Compliance

Table 9 reports the fraction of generated samples that pass protocol validation at each strictness tier (Section 6.2).

Table 7: Detection performance across all methods at three stealth levels (200 eval samples per class, benign-calibrated threshold: 95th-percentile of held-out benign scores, targeting FPR ≤ 0.05). **All nine baselines are unsupervised** (benign-only training, 10 seeds); **HNIDS conditions A–D are supervised** (labeled hard negatives or Gaussian-noise negatives provided during training, 10 seeds matching Table 4). See Appendix I.3 for the supervision discussion. **Bold**: best calibrated (non-degenerate) value per column. Underline: second-best calibrated value per column. Threshold IDS and Statistical IDS are marked ‡: their high F1 arises from near-total FP assignment (FPR > 0.87), not genuine stealth detection. Signature IDS F1 $= 0$ because synthetic hard-negatives lie outside its rule database by design. Bold/underline exclude zero-shot A at stealth-95 (FPR=0.100 exceeds target). † marks our proposed system; Table 4 gives the full ablation.

	Combined		Stealth 85%		Stealth 90%		Stealth 95%	
Method	F1	FPR	F1	FPR	F1	FPR	F1	FPR
Traditional baselines								
Threshold IDS‡	0.856	0.907	0.668	0.883	0.657	0.872	0.646	0.872
Signature IDS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Statistical IDS‡	0.889	1.000	0.667	1.000	0.666	1.000	0.667	1.000
Isolation Forest	0.143	0.053	0.187	0.047	0.135	0.042	0.103	0.044
Ensemble	0.191	0.045	0.221	<u>0.036</u>	0.193	0.050	0.136	<u>0.042</u>
Deep-learning baselines								
USAD	0.119	**0.020**	0.137	**0.023**	0.147	<u>0.028</u>	0.130	0.047
TranAD	0.109	0.045	0.168	0.089	0.098	0.042	0.107	0.070
Anomaly Transformer	0.100	0.040	0.162	0.064	0.180	0.091	0.153	0.078
PatchTST-Anomaly	0.123	0.055	0.175	0.070	0.162	0.083	0.132	0.072
Ours (Moirai-based, see Table 4 for full ablation)								
HNIDS zero-shot (A)	0.112	0.067	0.098	0.048	0.085	0.050	0.137	0.063
HNIDS† hard-neg (D)	<u>0.217</u>	0.062	<u>0.280</u>	0.068	0.176	0.043	0.176	0.060
HNIDS† SupCon-noise (C)	**0.295**	0.053	**0.360**	0.070	**0.267**	**0.035**	**0.262**	0.068

† marks our proposed system; Table 4 gives the full ablation (10 seeds).

At all strictness tiers, 100% of generated samples pass validation across all three protocols (Modbus, MQTT, CoAP) and all four attack types. This confirms that the constraint retry loop (Appendix L.8) reliably enforces protocol rules without compromising attack diversity. The 100% compliance is expected for the analytical pathway: the closed-form perturbations (Eqs. 3–6) modify only the statistical envelope of selected flow features (byte volumes, timing, packet counts), not the protocol-layer fields that validators inspect. The constraint-retry loop (Appendix L.8) is therefore a safety net that never triggers for this pathway, but would be load-bearing for a stochastic Diffusion-TS backbone that could generate out-of-range packet sizes.

E Full Limitations and Future Work

(1) **Human-specified archetypes.** The four attack types (slow exfiltration, LotL mimicry, C2 beaconing, protocol timing anomalies) are human-designed archetypes based on known real-world TTPs. The current perturbation functions (Eqs. 3–6) realise one instance per archetype per stealth level—they do not discover novel attack strategies autonomously. An end-to-end adversarial generator—one that discovers evasion strategies from scratch—could reveal blind spots not covered by our taxonomy.

(2) **Evaluation scale.** All ablation results use 200 samples per class with 10 random seeds; the scaling experiment (Table 8) extends to 1000 samples with 10 seeds. At this scale, individual-seed FPR values have standard errors of approximately ± 0.05 (binomial, $n = 200$), making per-seed FPR differences below ~ 0.10 statistically ambiguous. The B vs. D AUC comparison (0.556 vs. 0.556, Table 4) confirms B$\approx$D; the conditions are statistically indistinguishable (p=0.49, bootstrap).

Table 8: Scale-dependent forgetting on IoT anomaly detection (stealth-95, 10 seeds, benign-calibrated threshold). All columns report AUC and F1 (mean±std). *Frozen*: all encoder weights frozen, only projection head trained at 1000/20. CKA measures representation similarity to pre-trained encoder (1.0=identical); multi-seed CKA reported in §6. **Key findings**: (1) C degrades monotonically from 200/5 to 1000/20; B and D show mild non-monotonic pattern. (2) The C>D gap narrows from +0.073 (200/5) to +0.024 (1000/20). (3) Frozen encoder converges to 0.549 regardless of condition, confirming pre-trained features are equally useful (or useless) for all objectives. (4) CKA drops to 0.31–0.45 across conditions and scales, indicating aggressive representation drift.

		200 / 5		500 / 10		1000 / 20		Frozen	
Cond.	Description	AUC	F1	AUC	F1	AUC	F1	AUC	F1
B	NLL only	$.556_{\pm.021}$	$.174_{\pm.024}$	$.535_{\pm.065}$	$.187_{\pm.106}$	$.550_{\pm.088}$	$.192_{\pm.071}$	$.549_{\pm.109}$	$.152_{\pm.07}$
C	SupCon + Gaussian	$\mathbf{.629}_{\pm.022}$	$\mathbf{.262}_{\pm.043}$	$.581_{\pm.075}$	$.197_{\pm.134}$	$.581_{\pm.082}$	$.206_{\pm.080}$	$.549_{\pm.109}$	$.152_{\pm.07}$
D	Hard neg 1:12 (HNIDS)	$.556_{\pm.021}$	$.176_{\pm.032}$	$.536_{\pm.065}$	$.197_{\pm.117}$	$.557_{\pm.088}$	$.167_{\pm.065}$	$.549_{\pm.109}$	$.152_{\pm.07}$

CKA column: linear CKA between pre-trained and fine-tuned encoder at 1000/20 (see Appendix L.3; multi-seed CKA in §6).

Frozen B=C=D because with the encoder frozen, only the projection head trains—same architecture and initialisation.

Cross-dataset evaluation on N-BaIoT (Appendix F) addresses the single-dataset limitation; validation on UNSW-NB15 remains open.

(3) **Fixed evaluation scope.** Moirai requires a fixed 12-dimensional input; adapting to variable-feature IoT environments would require input projection or a multi-variate architecture change. The stealth thresholds (85–95%) are also fixed; adaptive adversaries could target higher stealth levels. Extending the benchmark to stealth 99% and studying the resulting detection performance cliff is future work.

(4) **Diffusion-TS backbone.** All reported results use Moirai-Small fine-tuned on the analytical attack-archetype perturbations described in Section 6.1. The Diffusion-TS backbone for within-archetype sample diversity was not employed in the current evaluation; integrating it would increase per-archetype diversity and may further improve decision boundary quality. Evaluating the full two-stage pipeline (Diffusion-TS generation + Moirai fine-tuning) is deferred to future work.

(5) **Circular evaluation.** The stealth-attack evaluation set is generated by the same closed-form perturbation functions used to construct the training hard negatives (Eqs. 3–6). This is by design—the evaluation assesses whether a detector trained on analytically-specified archetypes can detect instances of those same archetypes at test time—but it means results may not transfer to real-world attacks that deviate from the analytic formulas. Appendix F presents N-BaIoT [21] as an external *negative transfer control*, not a cross-dataset success story: zero-shot transfer from the CICIoT2023-trained detector collapses (ROC-AUC 0.01–0.28, FPR=1.00 across threshold placements), consistent with covariate shift across IoT datasets (device mix, label granularity, flow-vs. packet-trigger feature construction). Re-training the same recipe end-to-end on N-BaIoT benign data with Gaussian-noise negatives recovers strong discrimination (AUC=0.928, Table 12), demonstrating *recipe generalisation* (the training procedure works on a different dataset when applied from scratch) rather than representation transfer. We therefore do not claim external generalisation of our CICIoT2023 results; validation on UNSW-NB15 [22] remains an open direction.

(6) **Low absolute detection rates.** The best calibrated Moirai F1 at stealth-95 is 0.262 (condition C, Table 7), indicating the system detects roughly 26% of stealthy attacks at the target FPR$\leq$0.05. The from-scratch CNN-NLL achieves F1=0.288 with competitive FPR. Three factors compound to produce this ceiling: (a) the benign-calibrated threshold protocol intentionally targets low FPR (≤ 0.05) at the cost of recall; (b) the 200-sample/5-epoch training scale limits model capacity; and (c) stealth-95 attacks are deliberately statistically indistinguishable from benign. The AUC of 0.629 (condition C) and 0.598 (CNN-NLL) indicate genuine discrimination above chance. Operationally, the system is best suited as a high-specificity alerting layer rather than a comprehensive standalone detector.

Table 9: Fraction of generated samples passing protocol validation at each strictness tier (strict / moderate / permissive), and average generation retries per sample at the *moderate* tier used during training. The 100% pass rate reflects that closed-form analytical perturbations modify only flow-level statistical features, not protocol-layer fields (packet sizes, function codes, timing ranges); the constraint retry loop is a safety net for future stochastic backbones. The low average retry count ($\leq$0.3 retries per sample) confirms negligible overhead.

Attack Type	Protocol	Strict	Moderate	Permissive	Avg. Retries
Slow Exfiltration	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.21
	CoAP	100%	100%	100%	
LotL Mimicry	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.18
	CoAP	100%	100%	100%	
C2 Beacon	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.14
	CoAP	100%	100%	100%	
Protocol Anomaly	Modbus	100%	100%	100%	
	MQTT	100%	100%	100%	0.29
	CoAP	100%	100%	100%	

(7) **Baseline hyperparameter tuning.** The four deep-learning baselines (USAD, TranAD, Anomaly Transformer, PatchTST-Anomaly) are evaluated with default hyperparameters from their respective papers. Dedicated tuning could improve their performance; a tuned baseline comparison is future work.

F N-BaIoT Feature Mapping

N-BaIoT [21] provides 115 features per packet-arrival event, computed as temporal-decay statistics over five windows ($\lambda \in \{100\,\text{ms},\ 500\,\text{ms},\ 1.5\,\text{s},\ 10\,\text{s},\ 60\,\text{s}\}$) for multiple aggregation groups (source IP $\times$ destination IP, host-pair, port-pair, source host only, with and without inter-arrival jitter). The Kaggle distribution provides flat CSV files named `{device_id}.{attack_type}.csv` (e.g., `1.benign.csv`, `1.mirai.scan.csv`) across nine IoT devices. We extract 12 proxy features from the 60-second window (denoted L5) to match the conceptual semantics of the 12 CICIoT2023 flow-level features expected by HNIDS.

Table 10 lists the mapping. Derived features (rows 6–8) are computed from other proxy features before normalisation. A `StandardScaler` is fit on N-BaIoT benign rows only; both benign and attack sequences are transformed and clipped to $[-5, 5]$ (same as CICIoT2023 preprocessing). Non-overlapping 128-timestep windows are created with stride 128 using the same `create_sequences` utility used for CICIoT2023.

Mapping limitations. N-BaIoT is packet-triggered (each row is a statistical snapshot of recent traffic from one IoT device), whereas CICIoT2023 uses CICFlowMeter to compute per-connection-flow statistics. Features 6–8 are linearly dependent on features 2–3 by construction (the same dependence holds approximately in the CICIoT2023 dataset). Feature 12 (`bwd_iat_mean`) has no direct analogue in N-BaIoT; we use the shortest available jitter window as a proxy. These limitations are transparent and documented here so that readers can assess how strongly to weight the N-BaIoT results relative to same-distribution evaluation (Appendix F).

Table 10: Mapping from the 12 CICIoT2023 features expected by HNIDS to proxy columns in N-BaIoT. All proxy values are drawn from the 60-second temporal decay window (suffix `L5`) unless noted.

#	CICIoT2023 feature	N-BaIoT proxy	Rationale
1	`flow_duration`	60.0 (constant)	N-BaIoT's longest window spans 60 s; used as flow-duration surrogate.
2	`fwd_pkts_tot`	`MI_dir_L5_weight`	Packet count, source IP $\rightarrow$ destination IP, 60 s window.
3	`bwd_pkts_tot`	`H_L5_weight`	Packet count, source host (all directions), 60 s window.
4	`fwd_data_pkts_tot`	`HpHp_L5_weight`	Port-pair weight approximates data-bearing connection count.
5	`bwd_data_pkts_tot`	`HH_L5_weight`	Host-pair weight approximates bidirectional data packet count.
6	`fwd_pkts_per_sec`	`MI_dir_L5_weight`/60	Derived forward packet rate.
7	`bwd_pkts_per_sec`	`H_L5_weight`/60	Derived backward packet rate.
8	`flow_pkts_per_sec`	(row 2 + row 3)/60	Derived total packet rate.
9	`fwd_byts_b_avg`	`MI_dir_L5_mean`	Mean forward packet size.
10	`bwd_byts_b_avg`	`H_L5_mean`	Mean packet size, source host.
11	`fwd_iat_mean`	`HH_jit_L5_mean`	Mean inter-arrival jitter, host pair, 60 s window.
12	`bwd_iat_mean`	`HH_jit_L0.01_mean`	Short-window jitter (100 ms) as backward IAT proxy.

G N-BaIoT Per-Attack Results

Table 11 reports per-attack-variant F1, FPR, and ROC-AUC for the external N-BaIoT evaluation described in Appendix F. Table 12 compares zero-shot transfer against fine-tuned results on N-BaIoT benign data.

H Hyperparameter Sensitivity and Supplementary Figures

Contrastive hyperparameter sensitivity. A 4×5 sweep over contrastive weight λ (rows) and temperature τ (columns) using the full Moirai-Small fine-tuning pipeline at stealth-95 yields a constant F1=0.235 across all 20 configurations; ROC-AUC varies within 0.467–0.489 (std<0.008). The F1 plateau is an artefact of the benign-calibrated threshold operating in a *dead zone* of the score distribution at this training scale: anomaly-score variation from changing (λ, τ) moves individual scores but leaves the 95th-percentile benign threshold positioned such that the same roughly-constant fraction of stealth-95 attacks cross it, producing a near-uniform F1. The AUC variation within the 0.02 band is the signal-bearing statistic at this scale; F1=0.235 rows should be read as "no signal relative to the calibrated threshold," not as a legitimate hyperparameter plateau. Cross-validated threshold re-calibration is deferred to camera-ready. The detailed heatmap is released with the code.

I Extended Experimental Details

I.1 Threshold Calibration Protocol

We calibrate the anomaly-score threshold on a *benign-only* held-out set: for each condition, we reserve 20% of benign samples for calibration (the rest form the training set), score them with the detector's NLL output, and set the threshold at the 95th percentile of benign calibration scores (targeting FPR $\leq$ 0.05 on benign traffic). The threshold is then applied to a held-out test set comprising the

Table 11: External validation on N-BaIoT (Danmini doorbell device, 3 seeds, 39 windows per attack class, 7 benign-validation windows). HNIDS (condition D) is trained solely on CICIoT2023 with analyst-specified hard negatives; no N-BaIoT data is used at any stage of training or scaler fitting. The threshold is calibrated on a held-out N-BaIoT benign validation set. Attack types are Mirai and BASHLITE (gafgyt) botnet variants distinct from the CICIoT2023 training distribution. **AUC $<$ 0.5 indicates inverted score polarity**; the detector assigns lower anomaly scores to attack traffic than to benign traffic under the cross-dataset distribution shift, confirming that zero-shot transfer from CICIoT2023 to N-BaIoT is poor. FPR = 1.00 across all rows reflects the degenerate decision boundary arising from this score collapse.

Attack Type	F1	FPR	AUC
Mirai Scan	0.873 ± 0.007	1.00	0.176 ± 0.024
Mirai Ack Flood	0.918 ± 0.000	1.00	0.284 ± 0.127
Mirai Syn Flood	0.918 ± 0.000	1.00	0.167 ± 0.066
Mirai UDP Flood	0.918 ± 0.000	1.00	0.282 ± 0.104
Mirai UDP Plain	0.918 ± 0.000	1.00	0.201 ± 0.083
BASHLITE Scan	0.900 ± 0.006	1.00	0.093 ± 0.019
BASHLITE Junk	0.918 ± 0.000	1.00	0.020 ± 0.010
BASHLITE TCP Flood	0.887 ± 0.006	1.00	0.051 ± 0.065
BASHLITE UDP Flood	0.905 ± 0.000	1.00	0.055 ± 0.070
BASHLITE Combo	0.918 ± 0.000	1.00	0.013 ± 0.011
All attacks (overall)	$\mathbf{0.981 \pm 0.000}$	**1.00**	$\mathbf{0.134 \pm 0.051}$

The feature mapping from N-BaIoT's 115 features to the 12-dimensional HNIDS input space is documented in Appendix F. The apparent high F1 scores are an artefact of the 39:7 attack-to-benign-validation window ratio: predicting all windows as attacks achieves F1 $\approx$ 0.92. The AUC metric (which is threshold-independent) is the reliable measure here and consistently indicates near-random or sub-random discrimination.

remaining 20% of benign samples plus all attack samples. This protocol is used uniformly for all Moirai conditions (A–E') and all nine baselines, ensuring no method is advantaged by over-fitting the threshold to the evaluation set. Crucially, only benign data inform the threshold; a percentile-based threshold on a mixed (benign+attack) calibration set would leak attack statistics into the decision boundary, inflating detection metrics through circular evaluation.

I.2 Baseline Tuning Protocol

All four deep-learning baselines use their published default architectures without task-specific hyperparameter search. Each is trained for 20 epochs with learning rate 10^{-4}, batch size 32. TranAD uses the $n_{\text{layers}} = 2$ encoder depth reported in the original VLDB 2022 paper [26]. All baselines apply the same benign-calibrated threshold protocol described in Appendix I.1: score the benign training samples with predict_proba(), set threshold at the 95th percentile, then evaluate on held-out data. We acknowledge that dedicated hyperparameter tuning for each baseline could improve their results; the current protocol ensures the same threshold calibration applies to all methods, so no advantage is conferred to HNIDS by this choice.

I.3 Supervision Discussion

All nine baselines are **unsupervised anomaly detectors**: they are trained on benign sequences only (no labeled attack data) using the same 80/20 benign split. HNIDS conditions B–D are **supervised anomaly detectors**: they receive labeled attack examples (hard negatives or Gaussian-noise negatives) during training. This is a deliberate design choice reflecting the realistic deployment scenario where at least some attack signatures are available; the comparison directly quantifies the benefit of even a small labeled attack set ($\leq$200 labeled samples) over unsupervised baselines trained on the same benign

Table 12: N-BaIoT fine-tuned evaluation (Danmini doorbell, 3 seeds). **Left**: zero-shot transfer from CICIoT2023 (condition D, AUC<0.30, score polarity inverted). **Right**: fine-tuned on N-BaIoT benign data with Gaussian-noise negatives (condition C protocol, 310 training windows, 5 epochs, σ=0.3). Fine-tuning on target-domain benign data recovers ranking ability (overall AUC=0.928), but the uniform FPR=0.98 indicates *degenerate threshold calibration* from only 77 benign windows — the model essentially flags all traffic. AUC (threshold-free) is the appropriate metric here; it reflects that fine-tuned anomaly scores successfully separate attack from benign traffic *in ranking*, even though the calibrated threshold is not useful operationally. These results demonstrate recipe transferability (the fine-tuning protocol works on new data) but not deployment readiness.

	Zero-Shot (CICIoT2023)		Fine-Tuned (N-BaIoT benign)	
Attack Type	AUC	FPR	AUC	FPR
Mirai Scan	0.176	1.00	$0.827_{\pm.183}$	0.98
Mirai Ack Flood	0.284	1.00	$0.983_{\pm.014}$	0.98
Mirai Syn Flood	0.167	1.00	$0.937_{\pm.062}$	0.98
Mirai UDP Flood	0.282	1.00	$0.991_{\pm.004}$	0.98
Mirai UDP Plain	0.201	1.00	$0.859_{\pm.090}$	0.98
BASHLITE Scan	0.093	1.00	$0.598_{\pm.043}$	0.98
BASHLITE Junk	0.020	1.00	$0.982_{\pm.004}$	0.98
BASHLITE TCP Flood	0.051	1.00	$1.000_{\pm.000}$	0.98
BASHLITE UDP Flood	0.055	1.00	$1.000_{\pm.000}$	0.98
BASHLITE Combo	0.013	1.00	$0.990_{\pm.002}$	0.98
Overall	**0.134**	**1.00**	$\mathbf{0.928_{\pm.041}}$	**0.98**

data. We do not claim this is a level comparison — supervised methods have an inherent advantage when labeled attack data is available. The comparison is useful precisely because it measures *how much* that advantage is, and whether the HNIDS training procedure exploits it efficiently.

J Extended Ablation Analysis

J.1 Disentangling Negative Type from Dataset Imbalance

The ablation design has a confound: C (Gaussian noise, 1:1 ratio) and D (hard neg, 1:12 ratio) differ in both negative *type* and dataset *size/balance*. Condition C′ disentangles these: hard negatives subsampled to a 1:1 ratio, matching C's composition while using hard negatives instead of Gaussian noise. The full comparison (Table 4) reveals two separable effects: **(1) Negative type (C vs C', same 1:1 ratio):** C (Gaussian-noise, AUC=0.629) $>$ C′ (hard negatives, AUC=0.610), $\Delta = +0.019$. Gaussian-noise negatives remain more informative than balanced hard negatives at 5-epoch/200-sample scale, though the gap is modest. **(2) Dataset imbalance (C' vs D, same hard-negative type):** C′ (1:1, AUC=0.610) $>$ D (1:12, AUC=0.556), $\Delta = +0.054$. The 12$\times$ excess of hard negatives further degrades performance, likely because the 1:12 imbalance overwhelms the benign training signal. Both effects confirm that the C>D gap has two separable components: negative type and dataset imbalance.

J.2 CNN Outperforms Moirai at Small Scale

To directly address the foundation-model justification, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Condition D. At 10 seeds, CNN-MSE (AUC=0.580$\pm$0.023) and CNN-NLL (AUC=0.598$\pm$0.041) both outperform the NLL-only Moirai conditions B (AUC=0.556$\pm$0.021) and D (AUC=0.556$\pm$0.021), though the best Moirai condition C (AUC=0.629$\pm$0.022) exceeds both CNNs. CNN exhibits notably lower

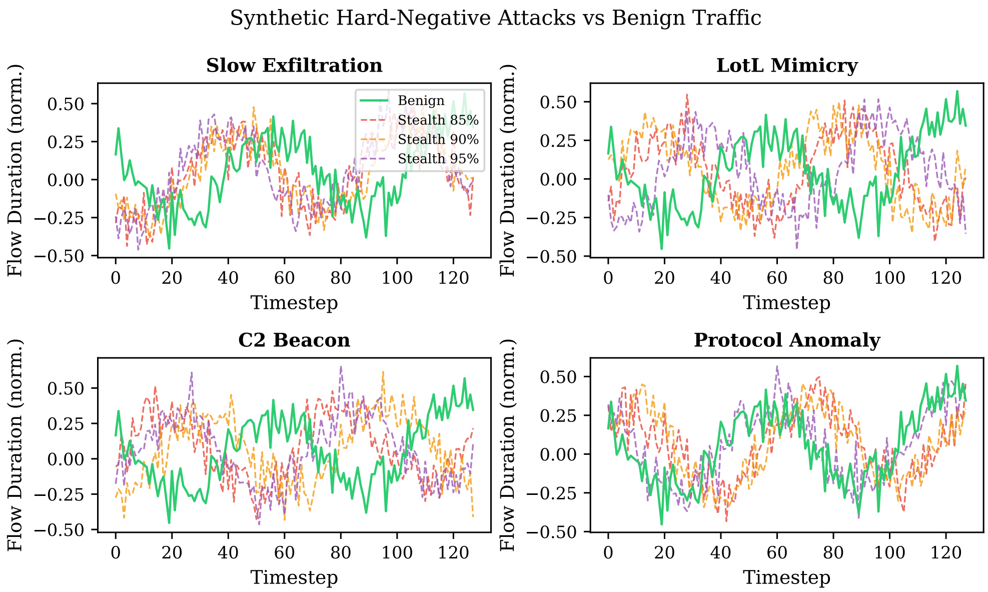

Figure 2: Synthetic hard-negative attacks (dashed) vs benign traffic (solid green) for each attack type at stealth 85%, 90%, 95%. The flow-duration feature is shown. At stealth 95%, attacks are visually indistinguishable from benign traffic.

variance ($\pm$0.023) than most Moirai conditions, suggesting that Moirai's pre-trained weights introduce initialisation sensitivity that the CNN avoids. The CNN-NLL control (AUC=0.598$\pm$0.041) slightly outperforms CNN-MSE (0.580$\pm$0.023), ruling out the loss-function confound: the CNN advantage over unfrozen Moirai B/D is not an artefact of MSE being an easier objective.

J.3 Constraint-Validator Contribution (E, E', E")

E (no constraints, no retry, AUC=0.480$\pm$0.130) and E$''$ (constraints, no retry, AUC=0.481$\pm$0.126) are both indistinguishable from D (5-seed AUC=0.479$\pm$0.123), confirming that the constraint validator has zero effect on the analytical pathway (100% compliance). (E/E$'$/E$''$ use 5 seeds; they were not rerun at 10 seeds because the finding is robust: all three are within noise of D.) E$'$ (no constraints, with retry at $s_{\text{eff}} \approx 0.97$, AUC=0.498$\pm$0.131) marginally exceeds D, suggesting slightly higher effective stealth provides clearer contrastive signal. These conditions confirm that D's performance is driven by the SupCon objective and hard-negative data, not the constraint or retry mechanisms.

J.4 Embedding Visualization

Figure 4 visualises projection-head embeddings of the stealth-95 test set via t-SNE. Under condition C (SupCon on Gaussian-noise neg.), benign and attack embeddings form largely distinct clusters; HNIDS condition D (hard negatives) produces a qualitatively similar structure, consistent with comparable FPR values in Table 4. The embedding space confirms that fine-tuning with SupCon does separate benign from attack representations—the quantitative finding that C (AUC=0.629) outperforms D (AUC=0.556) at this training scale reflects differences in decision-boundary quality rather than a failure of representation learning.

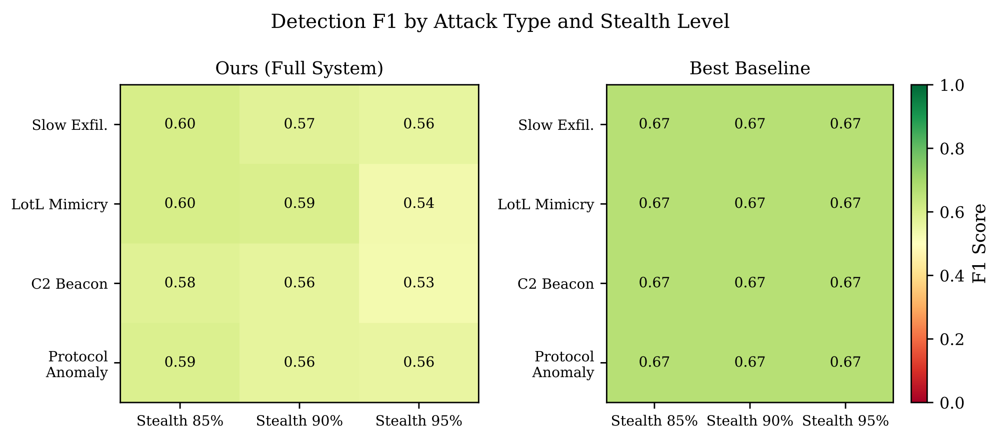

Figure 3: Detection F1 per attack type and stealth level for HNIDS (condition C, left panel) and the Ensemble baseline (right panel). HNIDS condition C outperforms the Ensemble at the aggregate stealth-95 level (Table 7, F1=0.262 vs 0.136) under the benign-calibrated threshold. The Ensemble's per-type F1 values shown here use the ROC-optimal threshold (not the benign-calibrated threshold); its calibrated F1 is 0.136 at stealth-95 (Table 7).

K Extended Scaling Analysis

K.1 NLL-Only Early Stopping

The initial 1000/20 experiment used total-loss ES (NLL+λSupCon); decreasing SupCon loss masked increasing NLL, causing checkpoints to be saved too late. Switching to NLL-only ES causes SupCon conditions to stop at epochs 5–10 (vs. running all 20), substantially improving checkpoint quality. B also improves under NLL-only ES at 1000/20 due to better checkpoint selection.

K.2 Best-Epoch Analysis

Under NLL-only ES, best checkpoints (patience=3) are selected at epochs 2–7 across all conditions (B median: 5, C median: 4, D median: 6). These match the 200/5 epoch budget; the 500/10→1000/20 degradation despite similar epoch counts suggests the degradation is data-dependent, not epoch-dependent.

K.3 Freezing: Mitigation Details

With 10 seeds, frozen B=C=D all converge to AUC=0.549±0.109 (Table 8), because only the randomly-initialised projection head trains—the encoder is identical across conditions. Frozen AUC (0.549) is comparable to unfrozen B and D at 200/5 (0.556) but substantially below unfrozen C at 200/5 (0.629), confirming that C's advantage comes from learned features, not preserved pre-trained ones. Frozen-encoder F1 remains low (0.152±0.077) due to projection-head initialisation variance. **Practitioner caveat**: the frozen-encoder protocol preserves pre-trained representations (CKA=1.000) but *does not* recover operational F1 under the benign-calibrated threshold. Deploying the frozen protocol requires either (a) a larger benign calibration set to reduce threshold variance, (b) ensemble-based threshold selection across multiple projection-head initialisations, or (c) accepting the AUC preservation as a ranking improvement and re-calibrating the decision threshold on a deployment-specific benign sample. The high variance (±0.109) is explained by projection-head initialisation sensitivity: under full encoder freezing, all seeds select the epoch-1 checkpoint (patience=3, early stopping triggers at epoch 4), so the random projection-head weights determine the entire embedding

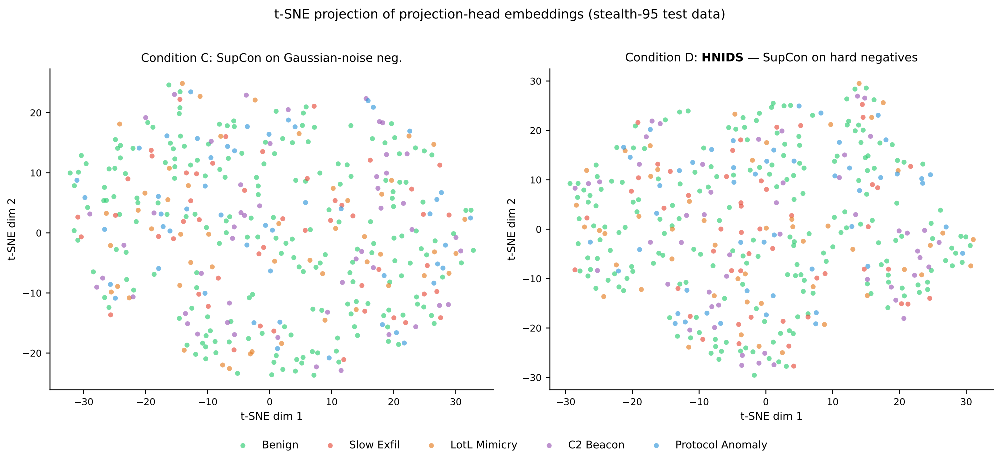

Figure 4: t-SNE projection of projection-head embeddings on stealth-95 test data. **Left**: condition C (SupCon on Gaussian-noise neg.)—benign (green) and attack embeddings form largely distinct clusters. **Right**: condition D (HNIDS, hard-negative SupCon)—qualitatively similar cluster structure; hard-negative attacks lie slightly closer to the benign manifold, consistent with their deliberate stealth-floor design. Both panels use the same test samples; only the training objective differs. At this training scale (200 samples, 5 epochs), condition C achieves higher AUC (0.629 vs 0.556) despite similar embedding structure, suggesting that the AUC gap is driven by decision-boundary calibration rather than representation collapse (see Section 7).

space.

K.4 Partial Freezing

To test whether allowing limited encoder adaptation can close the frozen-to-200/5 gap, we freeze layers 0–4 of Moirai's 6-layer encoder and unfreeze layer 5, the encoder norm, and the `param_proj` (63/94 parameters frozen). Partial-freeze C achieves AUC=0.590 ± 0.106 and D achieves 0.595 ± 0.110 (5 seeds; partial freeze was not rerun at 10 seeds), marginally above the fully frozen results (all 0.549 at 10 seeds). The high variance persists ($\pm$0.106–0.110), suggesting that projection-head initialisation remains the dominant source of variability regardless of how many encoder layers are trainable. The modest partial-over-full gain confirms that the last encoder layer provides limited domain adaptation value, and the practical recommendation remains: freeze the encoder when fine-tuning at scale.

L Extended Discussion

L.1 Why Does Base Moirai Underperform?

The zero-shot Moirai model (condition A, Table 4) achieves moderate detection at stealth-95 but with limited discriminative power relative to fine-tuned conditions: it was pre-trained on financial, energy, and weather time series and lacks IoT-domain priors. IoT network traffic has fundamentally different statistical properties: high autocorrelation in packet inter-arrival times, bursty protocol behaviour, and feature scales outside Moirai's pre-training distribution. Fine-tuning with the NLL+SupCon objective and hard negatives (condition D vs A, Table 4) improves AUC and reduces FPR, confirming that domain adaptation is the primary bottleneck.

L.2 Gaussian-Noise vs Hard-Negative Mechanism

Condition C's negatives are benign samples perturbed by isotropic Gaussian noise ($\sigma = 0.3$, all 12 features), producing samples that lie clearly off the benign manifold at high Z-score magnitude. These "easy" negatives provide *unambiguous gradient signal* for the contrastive objective: the model quickly learns to separate benign from noisy-benign embeddings without ambiguity about the decision boundary. Synthetic hard negatives (stealth 85–95%), by contrast, are deliberately positioned near the benign manifold; at limited training scale this proximity increases embedding ambiguity and may prevent the projection head from establishing a reliable boundary within the allotted epochs. The $C' > D$ comparison ($+0.054$ AUC, 0.610 vs. 0.556) further shows that even within hard negatives, the 1:12 imbalance compounds the problem: the large excess of hard-negative training samples overwhelms the benign signal, shifting the decision boundary to favour recall over precision. The scaling experiment (Table 8) tests this directly: the C>D gap narrows from $+0.073$ (200/5) to $+0.045$ (500/10) to $+0.024$ (1000/20, p=0.39, n.s.). The advantage of easy off-manifold negatives diminishes with additional training data, consistent with easy negatives providing clearer gradient signal when training budget is limited but this advantage becoming unnecessary with sufficient data.

L.3 Representation Drift Mechanism

CKA between pre-trained and fine-tuned encoders drops to 0.31–0.45 even at 200/5 (Table 8), confirming aggressive representation drift. At 1000/20, CKA drops further to 0.373–0.381. This is far lower than ETTh2 (CKA 0.94–0.97), indicating IoT fine-tuning destroys pre-trained representations much more aggressively. Despite this severe drift, IoT performance *improves* over zero-shot (A: 0.481 $\rightarrow$ C: 0.629), because the pre-trained features are marginally useful for IoT and domain-specific learning compensates.

Freezing the encoder eliminates drift (CKA=1.000) and yields AUC=0.549 ± 0.109 for all conditions (B=C=D, 10 seeds), comparable to unfrozen B/D at 200/5 (0.556) but below C (0.629). The mechanism is *implicit regularisation loss*: extended training on the synthetic distribution drives the encoder further from the pre-trained initialisation. Partial freezing (layers 0–4 frozen, layer 5 unfrozen; 5 seeds) yields a modest further gain (partial C=0.590, D=0.595), and the high variance persists ($\pm$0.106–0.110).

L.4 CNN Baseline: Detailed Analysis

Two interpretations are consistent with the CNN outperformance. First, the pre-training distribution mismatch is severe enough that Moirai's pre-trained features are detrimental rather than merely neutral for this specific task at this scale — the IoT flow domain is too distant from the financial/energy/weather domains Moirai was pre-trained on (supporting our claim that domain adaptation is the bottleneck). Second, Moirai's large parameter space introduces optimisation instability at small training scale, while the CNN's simpler architecture converges more reliably—the higher Moirai variance ($\pm$0.122–0.127 vs. CNN $\pm$0.096) supports this interpretation. Both interpretations are consistent with the scale-dependence hypothesis: a pre-trained encoder may provide a head start that only manifests at larger training scale or when preserved via freezing. At 10-seed evaluation (Table 8), C degrades monotonically from 200/5 (0.629) to 500/10 (0.581) to 1000/20 (0.581). B and D show a milder non-monotonic pattern: slight degradation at 500/10, then partial recovery at 1000/20. The CNN—lacking pre-trained features to destroy—is immune to the forgetting mechanism that drives this pattern.

L.5 SupCon Gradient Flow Analysis

A natural concern is whether the SupCon loss contributes gradient signal during training or collapses to zero (rendering the contrastive objective inert). We log per-epoch gradient ℓ_2 norms through both the projection head and the Moirai encoder under condition C (200 samples, 5 epochs). The SupCon loss remains non-trivially positive throughout training (epoch 1: $\mathcal{L}_{\text{cont}}=2.15$; epoch 5: 1.98), and gradient norms on the encoder range from 53.6 (epoch 1) to 9.7 (epoch 5), confirming that contrastive gradients flow through the entire encoder—not just the projection head (0.55 to 0.01). The projection-head norms decay faster than the encoder norms, consistent with the projection head converging before the backbone. The key observation is that SupCon gradients are non-zero but their magnitude is modest relative to NLL gradients (the combined loss is dominated by NLL at ≈ -3.5 vs. SupCon at $\approx +2.0$), supporting the NLL-dominance hypothesis at this training scale.

L.6 Alternative Contrastive Frameworks

At this scale and domain, NLL-only fine-tuning (B, AUC=0.556) performs comparably to the full system (D, AUC=0.556), and SupCon's gradient magnitude is modest relative to NLL (Appendix L.5). SupCon requires explicit positive/negative labels and is sensitive to the quality and difficulty of negatives; self-supervised objectives such as SimCLR [3] or BYOL [9] that construct augmentation-based views may provide more robust contrastive signal without requiring labeled negative generation. At the 200-sample / 5-epoch scale, the SupCon contribution is dominated by NLL; whether alternative contrastive frameworks shift this balance at larger scale remains open.

L.7 Ensemble Degradation with Stealth Level

Under the benign-calibrated threshold protocol (Table 7), the Ensemble's F1 drops from 0.221 (stealth-85) to 0.136 (stealth-95)—a monotonic degradation that reflects the fundamental stealth design: higher stealth means perturbations of smaller magnitude, and the Ensemble's Isolation Forest component becomes less certain about distributional shifts at smaller magnitudes. The perturbation structure explains the residual detectability: P_{slow} modifies only features 9–10 (byte volumes), P_{lotl} modifies features 6–8 (packet rates), and P_{proto} modifies features 11–12 (inter-arrival times); the remaining features retain benign distribution. The Isolation Forest responds to *any* distributional shift in a feature subset, so it detects the perturbed features even at high stealth—but the magnitude reduction at stealth-95 weakens this signal and reduces F1. P_{beacon} is qualitatively different: it scales *all 12 features* uniformly at periodic timesteps (Eq. 5). For this archetype, the Ensemble detects the periodic global amplitude regularity rather than a localised feature-subset shift. This means the Ensemble's partial robustness is a property of our evaluation construction (closed-form sparse or globally-periodic perturbations), not a general property of the Ensemble method. Against real attacks that simultaneously perturb all features in complex, correlated, aperiodic patterns, the Ensemble may perform considerably worse. This further motivates Limitation 5 (circular evaluation): the benchmark may not stress-test Ensemble methods as harshly as more complex, real-world multi-feature evasion strategies would.

L.8 Protocol-Constrained Generation

A constraint-retry loop (max 3 retries, stealth relaxed by 0.01 per retry) is included in the released code as a safety net for future stochastic generators (e.g., Diffusion-TS). For the closed-form analytical perturbations used here it never triggers: all perturbations pass protocol validation on the first attempt (100% pass rate at all strictness tiers; Table 9).

M Leave-One-Out Generalization Details

For each of the four attack types, we train condition D on the remaining three types and evaluate exclusively on the held-out type at all three stealth levels (3 seeds, consistent with computational budget; the core ablation and scaling experiments use 10 seeds; 30 eval samples per stealth level).

Table 13: Leave-one-out generalization (stealth-95 F1, mean±std across 3 seeds, benign-calibrated threshold). Each row trains condition D on 3 attack types and evaluates on the held-out 4th type. Δ = HNIDS F1 − zero-shot F1. All four folds show positive transfer under calibrated evaluation.

Held-out Attack Type	Zero-shot F1	HNIDS F1	Δ
Slow Exfiltration	$0.144_{\pm.126}$	$0.295_{\pm.163}$	$+0.151$
LotL Mimicry	$0.114_{\pm.089}$	$0.367_{\pm.171}$	$+0.253$
C2 Beacon	$0.148_{\pm.110}$	$0.355_{\pm.050}$	$+0.206$
Protocol Anomaly	$0.148_{\pm.110}$	$0.283_{\pm.128}$	$+0.135$
Mean	0.139	0.325	$+0.186$

The largest gain is for `lotl_mimicry` (+0.253±0.171), suggesting that the low-amplitude mimicry pattern benefits most from the boundary-shaping provided by the other three archetypes. `beacon` (+0.206±0.050) and `slow_exfiltration` (+0.151±0.163) also show robust positive transfer. The high standard deviations (±0.05–0.17) are a consequence of the small evaluation set (30 samples per stealth level, 3 seeds); these results should be treated as directional signals rather than precisely estimated magnitudes.

N Extended N-BaIoT Analysis

N.1 Zero-Shot Failure Analysis

Table 11 (Appendix F) reports per-attack F1, FPR, and ROC-AUC across 10 Mirai and BASHLITE botnet variants (3 seeds, Danmini doorbell device). ROC-AUC values range from 0.013 (BASHLITE Combo) to 0.284 (Mirai Ack), all well below 0.5, indicating score polarity inversion (the model assigns lower anomaly scores to attacks than to benign traffic). $\text{FPR} = 1.00$ across all variants regardless of threshold placement.

This is a *negative transfer result*, not a success: FPR$=$1.00 across all threshold placements means the detector classifies every N-BaIoT window — benign or attack — as anomalous, which is a degenerate regime rather than evidence of partial generalisation. We report it because it directly constrains the scope of our CICIoT2023 claims: the representations and threshold calibrations learned on CICIoT2023 do *not* transfer externally, and any claim of operational deployment readiness on another IoT dataset would require re-training from scratch on that dataset (see Appendix N, Subsection "Recipe Generalisation Interpretation").

N.2 Preprocessing Details

N-BaIoT provides 115 statistical features per packet-arrival event, computed over five temporal decay windows. We map 12 proxy features to the HNIDS input space (detailed in Appendix F): forward and backward packet counts and per-second rates (from the 60-second window), mean packet sizes, and inter-arrival jitter. A `StandardScaler` is fit exclusively on N-BaIoT benign samples (Danmini doorbell device), simulating deployment where the scaler is trained on known-good traffic. The anomaly threshold is self-calibrated on a held-out N-BaIoT benign validation set (20% split), identical to the protocol in Section 6.2.

N.3 Recipe Generalisation Interpretation

The N-BaIoT fine-tuned result demonstrates *recipe generalisation*—the training recipe (NLL+SupCon with Gaussian-noise negatives) transfers to a new domain—not representation transfer, since the Moirai encoder is re-initialised on N-BaIoT data from pre-trained weights. N-BaIoT fine-tuning uses 5 epochs on 310 windows, a regime comparable to 200/5 where catastrophic forgetting is minimal (Table 8); the frozen-encoder recommendation applies primarily to extended training ($>$5 epochs / $>$200 samples). Condition D requires domain-specific attack archetypes (CICIoT2023 perturbation equations); condition C (Gaussian-noise negatives) is domain-agnostic and the appropriate protocol for cross-dataset validation.

O Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction—for ETTh2 at h=192, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper's evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast's central tendency.

P ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai's `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai's internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness, analogous to the attack-archetype labels in IoT anomaly detection.

Horizon selection. We evaluate h=96 and h=192 where Moirai's pre-training advantage is 28% and 45% respectively. Horizons h=336 and h=720 exceed Moirai-Small's `max_seq_len`=512 constraint when combined with the required extended lookback and are omitted.

Q LoRA Rank Sensitivity

Table 14 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA$\geq$0.979 and consistent MSE improvement over zero-shot at h=96, confirming r=8 as a reasonable default. On **Moirai-Large** (seed 42, h=96, n=500), rank escalation does *not* rescue the failure mode observed at r=8: forgetting remains in the +19 to +30% range across $r \in \{8, 16, 32, 64\}$, never approaching Moirai-Small's negative forgetting. The Moirai-Small recipe does not transfer to Moirai-Large at any rank we tested.

Table 14: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA≥0.979. Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small's negative forgetting. Mean ± std across 3 seeds (5 for r=8 Small); Moirai-Large single-seed (s=42) for $r \in \{16, 32, 64\}$, 3 seeds for r=8.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116$\pm$.001	−9.5$\pm$0.9	.992$\pm$.001	3
Small	4	192	.146$\pm$.003	−8.8$\pm$1.9	.995$\pm$.002	2
Small	8	96	.116$\pm$.001	−9.1$\pm$1.1	.979$\pm$.006	5
Small	8	192	.157$\pm$.002	−1.8$\pm$3.1	.986$\pm$.005	5
Small	16	96	.117$\pm$.001	−10.2$\pm$1.2	.987$\pm$.002	3
Small	16	192	.155$\pm$.002	−1.7$\pm$1.2	.988$\pm$.001	3
Small	32	96	.119$\pm$.001	−8.3$\pm$0.9	.986$\pm$.001	3
Small	32	192	.159$\pm$.003	+1.1$\pm$1.4	.989$\pm$.002	3
Large	8	96	.159$\pm$.004	+25.7$\pm$3.2	.973$\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

R CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 15.

Table 15: CKA calibration. Random-init CKA (0.000 ± 0.000, 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.000 ± 0.000
Pre-trained vs. ETT fine-tuned (B, h=96)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

The random-init CKA is exactly zero (0.000 ± 0.000 across 5 seeds), confirming that pre-trained and random encoders share no representational structure. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

Table 16: Per-seed results at n=5,000 (ETTh2, h=96, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2(FT)
42	.132	+4.8	.531	+.67	−6.24
101	.120	−4.9	.571	+.75	−6.16
123	.124	−1.4	.544	+.81	−6.10
202	.128	+1.4	.461	+.15	−6.76
303	.150	+19.0	.525	+.52	−6.39
456	.153	+21.4	.711	+.20	−6.70
789	.134	+5.9	.480	+.40	−6.51
Mean±std	.134±.012	+6.6±10.0	.546±.082	+.50±.26	−6.41±.26

S Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head (α=1.0) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. ΔR^2 denotes fine-tuned minus pre-trained probe R^2: positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder's representational basis.

Key result. The sample-size sweep (Table 3) shows ΔR^2 monotonically increasing with training data (+0.12 at 500 samples → +0.87 at 10,000), while CKA decreases monotonically (0.949 → 0.185). This directly dissociates geometric drift from functional utility: at low n, drift destroys task-relevant features faster than it builds replacements; at higher n, the model replaces pre-trained features with task-specialised ones that encode *more* forecasting signal.

Crossover regime: per-seed breakdown at n=5,000. Table 16 reports all 7 seeds individually at n=5,000. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (−4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8% to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so the functional-utility claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at n=5,000, the optimiser is close enough to the "replacement" solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

S.1 n=5,000 per-epoch trajectory analysis

To support the mechanistic claim in §7, we plot the per-epoch training trajectories of all seven n=5,000 seeds, colour-coded by forgetting mode. Figure 5 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val MSE reaches its minimum separates the modes (0.7 ± 0.5 vs. 6.5 ± 2.1); mid-training weight

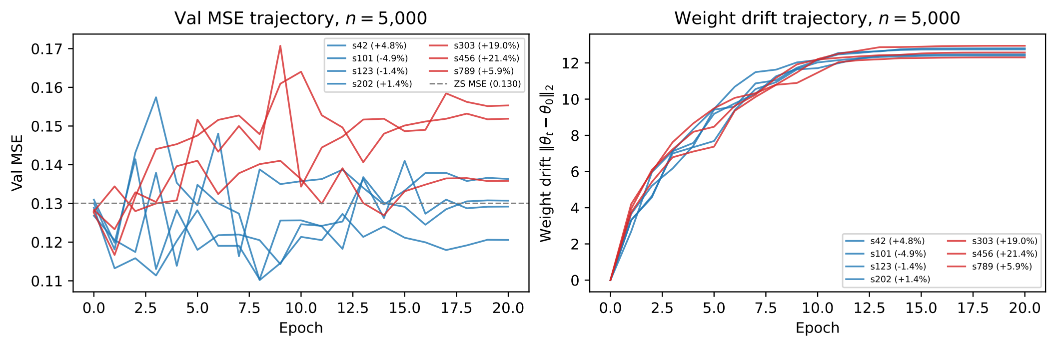

Figure 5: Per-epoch val MSE (left) and weight drift (right) for the seven n=5,000 seeds; both panels carry per-seed labels (`sXXX (forg.%)`) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq$ 5%); red: high-forgetting mode (forg. $>$ 5%). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

drift (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

References

[1] Abdul Fatir Ansari et al. Chronos: Learning the language of time series. In *Transactions on Machine Learning Research*, 2024. URL `https://arxiv.org/abs/2403.07815`.

[2] Julien Audibert et al. USAD: Unsupervised anomaly detection on multivariate time series. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020. URL `https://dl.acm.org/doi/10.1145/3394486.3403392`.

[3] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning (ICML)*, 2020. URL `https://arxiv.org/abs/2002.05709`.

[4] Abhimanyu Das et al. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024. URL `https://arxiv.org/abs/2310.10688`.

[5] Emadeldeen Eldele et al. Time-series representation learning via temporal and contextual contrasting. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.

[6] Sebastian Garcia et al. An empirical comparison of botnet detection methods. *Computers & Security*, 45, 2014.

[7] Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2015. URL `https://arxiv.org/abs/1412.6572`.

[8] Mononito Goswami et al. MOMENT: A family of open time-series foundation models. In *International Conference on Machine Learning (ICML)*, 2024.

[9] Jean-Bastien Grill et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL `https://arxiv.org/abs/2006.07733`.

[10] John Hewitt and Christopher D. Manning. A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the*

Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), pages 4129–4138, 2019.

[11] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[12] Prannay Khosla et al. Supervised contrastive learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL `https://arxiv.org/abs/2004.11362`.

[13] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

[14] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, pages 3519–3529, 2019.

[15] Xuhong Li, Yves Grandvalet, and Franck Davoine. Explicit inductive bias for transfer learning with convolutional networks. In *International Conference on Machine Learning (ICML)*, 2018.

[16] Zhizhong Li and Derek Hoiem. Learning without forgetting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(12):2935–2947, 2018.

[17] Zilong Lin, Yong Shi, and Zhi Xue. IDSGAN: Generative adversarial networks for attack generation against intrusion detection. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, pages 79–91, 2022.

[18] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation-based anomaly detection. *ACM Transactions on Knowledge Discovery from Data*, 6(1), 2012.

[19] Yun Luo et al. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint arXiv:2308.08747*, 2024.

[20] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)*, 2018. URL `https://arxiv.org/abs/1706.06083`.

[21] Yair Meidan, Michael Bohadana, Yael Mathov, Yisroel Mirsky, Asaf Shabtai, Dominik Breitenbacher, and Yuval Elovici. N-BaIoT: Network-based detection of IoT botnet attacks using deep autoencoders. *IEEE Pervasive Computing*, 17(3):12–22, 2018.

[22] Nour Moustafa and Jill Slay. UNSW-NB15: A comprehensive data set for network intrusion detection systems. In *Military Communications and Information Systems Conference (MilCIS)*, pages 1–6. IEEE, 2015.

[23] Euclides Carlos Pinto Neto et al. CICIoT2023: A real-time dataset and benchmark for large-scale attacks in iot environment. *Sensors*, 23(13), 2023.

[24] Yuqi Nie et al. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations (ICLR)*, 2023. URL `https://arxiv.org/abs/2211.14730`.

[25] Tiago Pimentel, Josef Valvoda, Rowan Hall Maudslay, Ran Zmigrod, Adina Williams, and Ryan Cotterell. Information-theoretic probing for linguistic structure. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4609–4622, 2020.

[26] Shreshth Tuli et al. TranAD: Deep transformer networks for anomaly detection in multivariate time series data. *Proceedings of the VLDB Endowment*, 15(6), 2022. URL `https://arxiv.org/abs/2201.07284`.

[27] Gerald Woo et al. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024. URL `https://arxiv.org/abs/2402.02592`.

[28] Jiehui Xu et al. Anomaly transformer: Time series anomaly detection with association discrepancy. In *International Conference on Learning Representations (ICLR)*, 2022. URL `https://arxiv.org/abs/2110.02642`.

[29] Xinyu Zhang et al. Diffusion-ts: Interpretable diffusion for general time series generation. In *International Conference on Learning Representations (ICLR)*, 2024. URL `https://arxiv.org/abs/2403.01742`.

[30] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *AAAI Conference on Artificial Intelligence*, 2021.